Chapter 3: Data Preprocessing

- Data Preprocessing: An Overview
- Data Cleaning
- Data Integration
- Data Reduction and Transformation
- Dimensionality Reduction
- Summary

Why Preprocess the Data? — Data Quality Issues

ทำไมถึงต้อง data ไม่ดี ทำไมต้อง data ครบ

- Measures for data quality: A multidimensional view
  - Accuracy: correct or wrong, accurate or not
  - Completeness: not recorded, unavailable, …
  - Consistency: some modified but some not, dangling, …
  - Timeliness: timely update? ← data update ไหม.
  - Believability: how trustable the data are correct? ← น่าเชื่อถือแค่ไหน
  - Interpretability: how easily the data can be understood? ↑ เข้าใจได้ง่าย มากน้อยแค่ไหน

Data Cleaning

- Data in the Real World Is Dirty: Lots of potentially incorrect data, e.g., instrument faulty, human or computer error, and transmission error
  - Incomplete: lacking attribute values, lacking certain attributes of interest, or containing only aggregate data
    - e.g., *Occupation* = " " (missing data)
  - Noisy: containing noise, errors, or outliers
    - e.g., *Salary* = "−10" (an error)
  - Inconsistent: containing discrepancies in codes or names, e.g.,
    - *Age* = "42", *Birthday* = "03/07/2010"
    - Was rating "1, 2, 3", now rating "A, B, C"
    - discrepancy between duplicate records
  - Intentional (e.g., *disguised missing* data)
    - Jan. 1 as everyone's birthday?

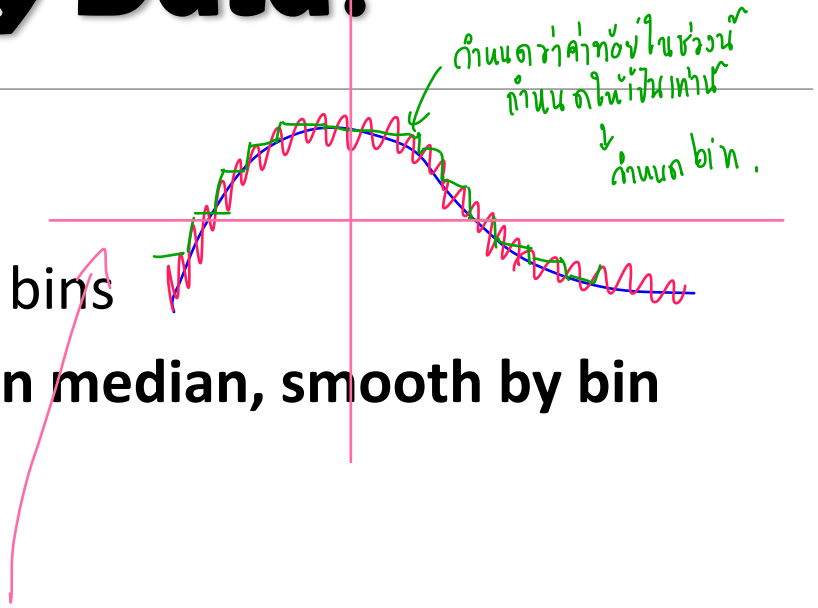
Incomplete (Missing) Data

- Data is not always available
  - E.g., many tuples have no recorded value for several attributes, such as customer income in sales data
- Missing data may be due to
  - Equipment malfunction
  - Inconsistent with other recorded data and thus deleted
  - Data were not entered due to misunderstanding
  - Certain data may not be considered important at the time of entry
  - Did not register history or changes of the data
- Missing data may need to be inferred

How to Handle Missing Data?

- Ignore the tuple: usually done when class label is missing (when doing classification)—not effective when the % of missing values per attribute varies considerably ลบทิ้ง → มีประโยชน์ เวลาลบแล้ว data หายไปไม่กี่ %
- Fill in the missing value manually: tedious + infeasible? เติม manual, ถ้าเติมเอง → ต้องรู้จัก data ดี ว่า data ควรเป็นอะไร
- Fill in it automatically with เติมไม่มนรู้ว่าเป็น missing. → ลองไปดูค่าตามกลุ่มดีๆ
  - a global constant : e.g., "unknown", a new class?!
  - the attribute mean → เติมค่ากลาง → (ค่าเฉลี่ย)
  - the attribute mean for all samples belonging to the same class: smarter ↑ ค่าเฉลี่ย ตาม class
  - **the most probable value: inference-based such as Bayesian formula or decision tree** ↳ ทำนาย missing โดยใช้ Model

How to Handle Noisy Data?



- Binning → เลือกค่า
  - First sort data and partition into (equal-frequency) bins
  - Then one can **smooth by bin means, smooth by bin median, smooth by bin boundaries**, etc.
- Regression → เอา function มา fit กัน → ทำให้มัน smooth
  - Smooth by fitting the data into regression functions
- Clustering → คล้าย binning → มีการจัด class ว่า data ตัวไหนควรอยู่ใน class นี้
  - Detect and remove outliers
- Semi-supervised: Combined computer and human inspection
  - Detect suspicious values and check by human (e.g., deal with possible outliers)

มีคนเข้ามาช่วยบอกว่าค่าไหนจะเป็นปัญหาบ้าง → ได้ data ที่ smooth
→ ได้ outlier. → outlier จริงๆ / outlier ที่ไม่สำคัญ

Data Cleaning as a Process

- **Data discrepancy detection**
  - Use metadata (e.g., domain, range, dependency, distribution)
  - Check field overloading → ค่าเกินจากที่ควรจะเป็นไหม
  - Check uniqueness rule, consecutive rule and null rule ← ทำไม data มันไม่ซ้ำกันเลย (บางอันอาจจะเปลกๆ ต้องเช็ค)
  - Use commercial tools
    - Data scrubbing: use simple domain knowledge (e.g., postal code, spell-check) to detect errors and make corrections
    - Data auditing: by analyzing data to discover rules and relationship to detect violators (e.g., correlation and clustering to find outliers)
- **Data migration and integration**
  - Data migration tools: allow transformations to be specified
  - ETL (Extraction/Transformation/Loading) tools: allow users to specify transformations through a graphical user interface
- Integration of the two processes
  - Iterative and interactive (e.g., Potter's Wheels)

extract transform loading.

Data Integration

- Data integration
  - Combining data from multiple sources into a coherent store
- Schema integration: e.g., A.cust-id ≡ B.cust-#
  - Integrate metadata from different sources
- **Entity identification:**
  - Identify real world entities from multiple data sources, e.g., Bill Clinton = William Clinton
- Detecting and resolving data value conflicts
  - For the same real world entity, attribute values from different sources are different
  - Possible reasons: different representations, different scales, e.g., metric vs. British units

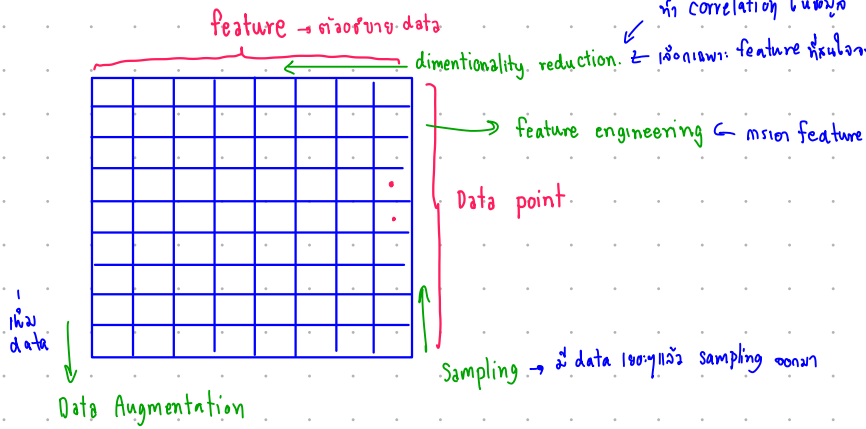
Handling Redundancy in Data Integration

- Redundant data occur often when integration of multiple databases
  - *Object identification*: The same attribute or object may have different names in different databases
  - *Derivable data:* One attribute may be a "derived" attribute in another table, e.g., annual revenue
- **Redundant attributes may be able to be detected by *correlation analysis* and *covariance analysis***
- Careful integration of the data from multiple sources may help reduce/avoid redundancies and inconsistencies and improve mining speed and quality

Dimensionality Reduction Techniques

- Dimensionality reduction methodologies
  - **Feature selection**: Find a subset of the original variables (or features, attributes)
  - **Feature extraction**: Transform the data in the high-dimensional space to a space of fewer dimensions
- Some typical dimensionality methods
  - Principal Component Analysis
  - Supervised and nonlinear techniques
    - Feature subset selection
    - Feature creation

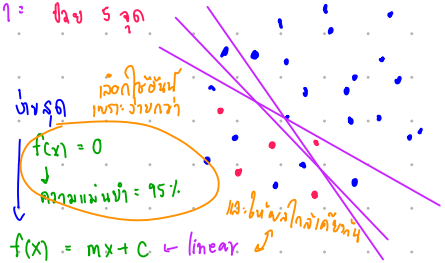
Dimentionality ~~Reduction~~ Modification



Majority = ข้อมูลกลุ่มใหญ่
Minority = ข้อมูลกลุ่มเล็ก

imbalance data

0 = คนปกติ 95 จุด
1 = ป่วย 5 จุด



ข้อดี 1 ทำ linear.

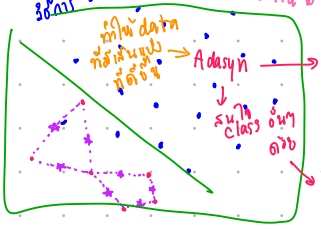
หน้าตาของ model ก็คือมีเส้นแบ่ง class

Occam Razor → ถ้า 2 อย่าง ทำงานใกล้เคียงกัน
จะเลือกอันที่ง่ายกว่าเสมอ

ถ้า data imbalance ก็ทำ data ให้เท่ากัน

- ลบตัวออก (under sampling)
- over sampling. (ลบตัวที่น้อย → duplicate)
- data Augmentation (เทคนิค SMOTE)

วิธีการ SMOTE แต่มีปัญหา คือ data ย้ายไปเส้นตรง → กระจุกตัว ไม่ล้อม ขาย (ไม่ดีอยู่ดี)

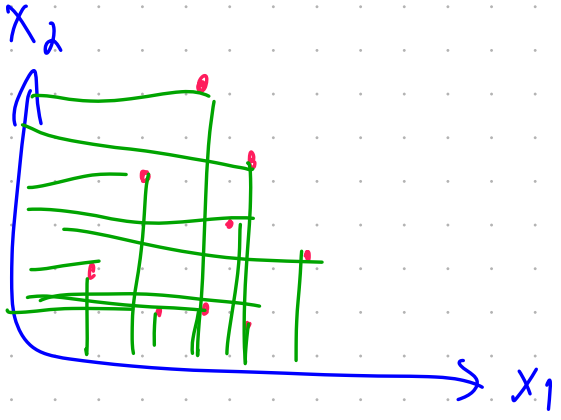


- พิจารณา weight boundary
  - แก้ไขโดยการ minority
  - ① เพิ่มข้อมูลประเภทมาหาของตัวเอง
  - ② (ลบข้อมูล Magority (ข้อมูลกลุ่มใหญ่))

ให้ความสำคัญกับตัวที่มีเพื่อนข้างเป็น Majority มาก เป็นพิเศษ.

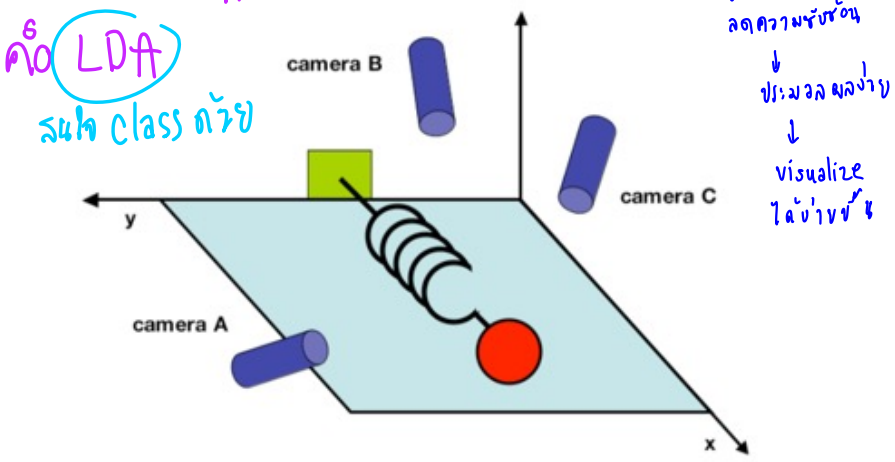
ทำไงดีจะสร้าง data มันออกมาจากสมการดี

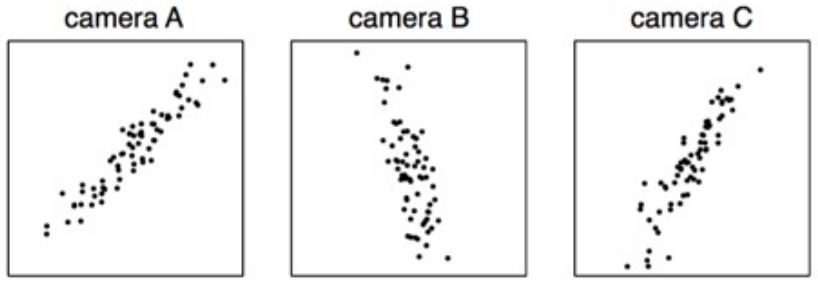
paper ที่ทำให้จุดมันออกมาอย่างนั้น



Principal Component Analysis (PCA)

- PCA: A statistical procedure that uses an orthogonal transformation to convert a set of observations of possibly correlated variables into a set of values of linearly uncorrelated variables called ***principal components***
- The original data are projected onto a much smaller space, resulting in dimensionality reduction
- Method: Find the eigenvectors of the covariance matrix, and these eigenvectors define the new space

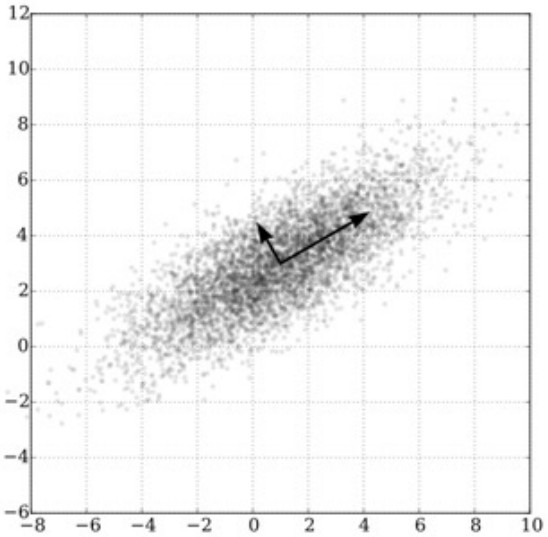




Ball travels in a straight line. Data from three cameras contain much redundancy

Principal Component Analysis (Method)

- Given *N* data vectors from *n*-dimensions, find $k \leq n$ orthogonal vectors (*principal components*) best used to represent data
  - Normalize input data: Each attribute falls within the same range
  - Compute *k* orthonormal (unit) vectors, i.e., *principal components*
  - Each input data (vector) is a linear combination of the *k* principal component vectors
  - The principal components are sorted in order of decreasing "significance" or strength
  - Since the components are sorted, the size of the data can be reduced by eliminating the *weak components*, i.e., those with low variance (i.e., using the strongest principal components, to reconstruct a good approximation of the original data)
- Works for numeric data only



Ack. Wikipedia: Principal Component Analysis